Quantifying the Annuity Puzzle: A Unified Lifecycle Decomposition*

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Abstract

Yaari (1965) showed that a consumer with uncertain lifetime and no bequest motive should fully annuitize under actuarially fair pricing. Observed voluntary annuity ownership among US retirees is only 3–6%. I build a lifecycle model nesting eleven channels proposed in the annuity-puzzle literature and quantify their joint contribution using exact Shapley values computed over all 2,048 channel subsets. The model layers in three blocks: seven rational channels (Social Security, bequest motives, medical expenditure risk, health-mortality correlation, survival pessimism, pricing loads, inflation erosion), two preference-based channels (age-varying consumption needs, state-dependent utility), and two behavioral channels that operate at distinct decision moments and point in opposite directions: source-dependent utility (Force A; Blanchett and Finke, 2024, 2025) raises annuitization by converting portfolio wealth into "spendable income," while a narrow-framing purchase penalty (Force B; Barberis and Huang, 2009; Tversky and Kahneman, 1992) suppresses it through loss aversion over the unrecouped premium until breakeven.

*Generative AI tools were used to assist with code development, manuscript preparation, and editing. All analysis, interpretation, and conclusions are the sole responsibility of the author. Replication code and data are available at `https://github.com/DerekTharp/annuity-puzzle-model`.

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Calibration of the narrow-framing penalty ψ_{purchase} is anchored to the UK 2015 pension-freedoms reform, the cleanest available natural experiment isolating choice architecture from price and product. Across alternative single-moment SMM specifications drawn from this reform, $\psi_{\text{purchase}} \in [0.0142, 0.0281]$ and the corresponding predicted US voluntary ownership ranges from 2.2% to 22.0%, a bracket that contains the observed HRS rate of 3.4%. The headline production point estimate of 17.4% uses the midpoint anchor (Anchor C-mid: 60 pp behavioral residual after correcting for the simultaneous tax-rule change). The remaining gap between the production point and observed ownership is informative about US-specific factors not captured in the UK calibration (advisor incentives, distribution channel frictions, and product complexity), rather than evidence against the model's structural account.

Under modern annuity pricing (MWR = 0.87, Wettstein et al., 2021), the seven rational channels reduce predicted ownership from a frictionless population benchmark of 41.4% to 35.6%; the preference channels bring the prediction to 22.3%; activating Force A alone raises it to 77.2%; activating Force B at the midpoint anchor yields the full eleven-channel point estimate of 17.4%. Both behavioral forces are individually large in absolute magnitude (Force A roughly +55 pp, Force B roughly −60 pp at the midpoint anchor) but partly offset, and they enter the exact Shapley decomposition with opposite signs (-36.7 vs. 28.3 pp), confirming that they are empirically distinct mechanisms rather than redundant parameterizations of the same wedge. The model's implied default-vs-opt-in elasticity at the midpoint anchor (59.7 pp) is consistent with the Goda-Manchester-Sojourner federal Thrift Savings Plan natural experiment, which was not used in calibration. Policy counterfactuals show that supply-side reform (group pricing) and demand-side reform (default annuitization) operate on distinct margins and are quantitatively comparable as policy levers.

JEL Codes: D14, D15, D91, G22, H55, J14

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1 Introduction

Yaari (1965) established one of the sharpest predictions in consumer theory: an individual facing uncertain lifetime, with no bequest motive and access to actuarially fair annuities, should convert all wealth into a life annuity. The mortality credit—the return premium financed by redistributing wealth from decedents to survivors—dominates any asset with the same underlying return. Davidoff et al. (2005) extended this result to incomplete markets, showing that partial annuitization remains optimal under a wide range of conditions. Yet among single US retirees aged 65–69, voluntary annuity ownership is approximately 3–6% (Lockwood, 2012; Dushi and Webb, 2004; Poterba et al., 2011). Six decades of research have proposed partial explanations for this gap, but no single paper has nested all the major channels in a unified framework and measured their joint quantitative effect.

This paper provides a calibrated structural accounting exercise for that gap. I build a lifecycle model incorporating eleven channels organized in three layers. The seven *rational* channels are pre-existing Social Security annuitization, bequest motives, medical expenditure risk, the health-mortality correlation identified by Reichling and Smetters (2015), subjective survival pessimism (O'Dea and Sturrock, 2023), pricing loads, and inflation erosion. The two *preference* channels are age-varying consumption needs (Aguiar and Hurst, 2013) and state-dependent utility (Finkelstein et al., 2013). The two *behavioral* channels operate at distinct decision moments and point in opposite directions: source-dependent utility (Force A) captures the post-purchase "license to spend" that retirees apply to guaranteed income flows but not portfolio assets (Shefrin and Thaler, 1988; Thaler, 1999; Blanchett and Finke, 2024, 2025); a narrow-framing purchase penalty (Force B) captures the at-purchase loss aversion over the unrecouped premium until breakeven (Tversky and Kahneman, 1992; Barberis and Huang, 2009; Brown et al., 2008; Chalmers and Reuter, 2012). All parameters are drawn from published sources; both behavioral parameters are identified out of sample from preference-elicitation experiments and quasi-experimental natural variation in default architecture, separate from the prediction target. The paper's contribution is the unified frame-

work, the introduction of age-varying consumption needs as a missing channel in the prior multi-channel literature, the formal incorporation of two countervailing behavioral channels each calibrated from independent moments, and an exact Shapley decomposition over all 2,048 channel subsets that attributes the demand reduction without order dependence.

The results have a three-layer structure. First, the seven standard rational channels reduce predicted ownership from a frictionless participation benchmark of 41.4% to 35.6% under modern annuity pricing (MWR = 0.87). That is a substantial reduction, but it still overpredicts observed ownership by a wide margin, echoing the residual gap in Pashchenko (2013). Second, adding age-varying consumption needs and state-dependent utility—the two preference channels—lowers predicted ownership to 22.3%. Third, adding the two behavioral channels produces a countervailing pattern: source-dependent utility raises predicted ownership to 77.2% by unlocking the "license to spend" on annuity income, while the narrow-framing purchase penalty subtracts even more demand. The headline production point estimate (Force B at the midpoint anchor of the UK 2015 behavioral residual) is 17.4%; the bracket across alternative single-moment SMM specifications, all anchored to the same UK reform, runs from 2.2% (anchor B-low: raw 75 pp UK ownership drop) through 11.8% (UK 65 pp behavioral) and 17.4% (UK 60 pp behavioral, the production point) to 22.0% (UK 55 pp behavioral). The bracket contains the observed HRS rate of 3.4%, so the model dissolves the puzzle in the sense that observed ownership lies within the empirically defensible range of behavioral elasticity. Each behavioral force is individually large in absolute magnitude (Force A roughly +55 pp, Force B roughly −60 pp at the midpoint anchor) but they partly offset. None of the three layers depends on tuning a free parameter to US ownership: λ_W is set ex ante from the Blanchett–Finke spending differential, and ψ_{purchase} is calibrated entirely from the UK natural experiment.

Relative to prior multi-channel papers, the present model broadens the channel set and changes the attribution. Pashchenko (2013) jointly considered Social Security, bequests, minimum purchase requirements, and pricing frictions, but not the Reichling–Smetters health-

survival mechanism, subjective survival pessimism, or age-varying needs. Peijnenburg et al. (2016) emphasized incomplete markets and background risk, concluding that the annuity puzzle remains unresolved in their framework. This paper shows that pricing, correlated health and mortality risk, and late-life expenditure decline are quantitatively central once evaluated in a common model.

The exact Shapley decomposition is the paper's preferred attribution result. Among demand-suppressing channels, the narrow-framing purchase penalty (Force B) has the largest Shapley value (28.3 pp, 118% of the full ownership reduction), followed by pricing loads (24.7 pp, 103%), the Reichling–Smetters health-mortality correlation (13.2 pp, 55%), age-varying consumption needs (7.3 pp, 30%), survival pessimism (6.5 pp, 27%), medical expenditure risk (5.5 pp, 23%), and inflation erosion (1.6 pp, 6%). Two channels enter with negative Shapley values, reflecting their role as ownership boosters: source-dependent utility (Force A, -36.7 pp, -153%) and Social Security (-28.5 pp, -119%). The two behavioral channels enter with opposite-sign Shapley values—the narrow-framing penalty as a suppressor and source-dependent utility as a booster—confirming empirically that they are not redundant parameterizations of a single behavioral mechanism. State-dependent utility and bequest motives both contribute negligibly (1.1 pp and 1.0 pp respectively), entering the paper as completeness checks rather than main drivers.

The policy and welfare results extend beyond the ownership match. Group annuity pricing (MWR = 0.90) raises predicted ownership from 17.4% to 26.2%, while removing the narrow-framing penalty entirely (default architecture; $\psi_{\text{purchase}} = 0$) raises it to 77.2%—a default-vs-opt-in elasticity consistent with the UK 2015 reform and the Goda-Manchester-Sojourner federal Thrift Savings Plan natural experiment, neither used in calibration. Social Security cut counterfactuals reveal that a 23% benefit cut—the projected consequence of trust fund exhaustion around 2033—raises private annuity demand to 57.5%, but the largest cuts eventually reduce demand by removing the income floor that makes private annuitization feasible at all. Under current pricing and the production calibration, the welfare gain

from annuity market access is small across the wealth distribution because most calibrated households optimally do not annuitize, so access is approximately neutral; policy interventions that change the optimal choice (group pricing, default architecture) generate the larger welfare gains.

These results shift the question away from whether retirees irrationally fail to annuitize and toward a more policy-relevant one: which channels suppress annuity demand most, and which market reforms would make annuitization welfare-improving for particular households?

Section 2 specifies the model. Section 3 describes the calibration. Section 4 presents the decomposition, counterfactuals, and welfare analysis. Section 5 reports sensitivity checks. Section 6 discusses limitations. Section 7 concludes.

2 Model

2.1 Environment

Time is discrete. An individual enters at age 65 (period $t = 1$) and may survive to a maximum age of 110 (period $T = 46$). At $t = 1$, the individual makes a single irreversible decision: what fraction $\alpha \in [0, 1]$ of initial liquid wealth W_0 to convert into a single premium immediate annuity (SPIA). The remaining wealth $(1-\alpha)W_0$ stays liquid. In each subsequent period, the individual chooses consumption c_t from available resources.

The state at each period t is characterized by four variables: liquid wealth $W_t \geq 0$, annuity income A (fixed after the age-65 purchase), health status $H_t \in \{1, 2, 3\}$ (Good, Fair, Poor), and age t. Social Security income $\mathrm{SS}(t)$ is received exogenously in each period and is not a decision variable.

2.2 Preferences

The individual maximizes expected discounted lifetime utility. Flow utility over consumption takes the constant relative risk aversion (CRRA) form:

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \quad \gamma > 0, \quad \gamma \neq 1. \tag{1}$$

The coefficient of relative risk aversion γ governs both risk aversion and the elasticity of intertemporal substitution.

Bequest utility follows the De Nardi–French–Jones specification, which nests bequests as a luxury good:

$$v(b) = \theta \frac{(b+\kappa)^{1-\gamma}}{1-\gamma}, \tag{2}$$

where b is bequeathable wealth (liquid wealth at death; annuity income ceases), $\theta \geq 0$ controls bequest intensity, and $\kappa > 0$ is a shifter that determines the degree to which bequests are a luxury good (De Nardi, 2004). When κ is large relative to typical wealth levels, the marginal utility of bequests is nearly flat for low-wealth individuals and declines steeply for the wealthy. This specification, estimated by Lockwood (2012) using Health and Retirement Study (HRS) data, captures the empirical pattern that bequest motives are concentrated among high-wealth households.

The parameters θ and κ must be interpreted jointly. With the DFJ luxury-good calibration ($\theta = 56.96$, $\kappa = \$272{,}628$), an individual with \$50,000 in wealth faces negligible marginal bequest utility, while an individual with \$1,000,000 faces strong resistance to annuitizing. Combining this θ with a small κ (e.g., \$10, the homothetic specification) produces pathological results: extreme marginal bequest utility near $b = 0$ acts as a precautionary buffer rather than a genuine bequest motive, anomalously *increasing* predicted ownership above the no-bequest case.[1] The DFJ estimates from Lockwood (2012) are jointly calibrated to

[1] At $\kappa = \$10$, marginal bequest utility near $b = 0$ is $\theta \cdot 10^{-\gamma}$, which is extreme relative to marginal consumption utility. Individuals avoid annuitizing not to protect bequests, but to avoid near-zero-wealth states where $v'(b)$ diverges. See Appendix C for details.

HRS bequest data; using one parameter with the other replaced is not a valid counterfactual.

The bequest channel's isolated effect on ownership is modest under the DFJ specification (100% retention rate in the sequential decomposition) because the luxury-good κ makes the bequest motive nearly irrelevant for the median retiree in the HRS sample. The 100% retention rate in the sequential decomposition reflects the ordering: bequests are added when ownership is already 100% (steps 0–1 include only SS). In the full model, removing bequests raises ownership from 35.6% to 19.3% (Table 8), because bequests interact with pricing loads and inflation to suppress demand for agents near the annuitization margin. The Shapley decomposition, which averages over all orderings, assigns bequests a value of 1.0 pp (4% share).

The individual discounts future utility at rate β per period.

2.3 Health dynamics and mortality

Health follows a first-order Markov process with three states: Good ($H = 1$), Fair ($H = 2$), and Poor ($H = 3$). Transition probabilities are age-dependent and calibrated to HRS panel data following De Nardi et al. (2010):

$$\Pr(H_{t+1} = h' \mid H_t = h, \text{age} = t) = \pi(h, h', t). \tag{3}$$

Mortality depends on health through a proportional hazard specification. Baseline survival probabilities $s_{\text{base}}(t)$ come from the SSA period life table used by Lockwood (2012). Health adjusts survival through:

$$s(t, H) = s_{\text{base}}(t)^{\mu(H,t)}, \tag{4}$$

where $\mu(H, t)$ is a health- and age-specific hazard multiplier with $\mu(\text{Good}, t) < 1 < \mu(\text{Poor}, t)$ and $\mu(\text{Fair}, t) = 1$. This specification maps directly to proportional hazards: if the baseline hazard is $\lambda_{\text{base}}(t) = -\ln s_{\text{base}}(t)$, then the adjusted hazard is $\mu(H, t) \cdot \lambda_{\text{base}}(t)$.

The baseline specification uses constant hazard multipliers $\mu = [0.50, 1.0, 3.0]$ for Good, Fair, and Poor health respectively, chosen as a central tendency across age bands from the HRS data. The health-mortality gradient compresses with age: the Poor-to-Fair hazard ratio falls from 3.29 at ages 65–74 to 2.77 at 75–84 to 1.82 at 85+. Section 5 reports results across the empirical range of multiplier specifications.

The age-varying multipliers are the mechanism underlying the Reichling and Smetters (2015) result. A negative health shock simultaneously increases mortality ($\mu(\text{Poor}, t) > 1$ reduces expected remaining annuity payments) and, through the medical expenditure process described below, increases the need for liquid wealth. The annuity becomes "valuation-risky": it loses value in precisely the states where liquidity is most valuable. The effect is strongest at younger ages where the gradient is widest, which matters because the annuitization decision is made at age 65.

2.4 Subjective survival beliefs

O'Dea and Sturrock (2023) document that individuals aged 65–69 underestimate their survival probability at age 75 by approximately 15 percentage points relative to actuarial tables (subjective Pr(survive to 75) ≈ 0.71 versus actuarial 0.86). This pessimism reduces the perceived value of annuities by overweighting the probability of dying before recovering the premium.

The model allows the agent's subjective survival probabilities to differ from the objective rates used by the insurer to price annuities:

$$s^{\text{subj}}(t, H) = \psi \cdot s(t, H), \quad \psi \in (0, 1]. \tag{5}$$

The parameter ψ scales each one-year survival probability downward. The annuity is priced using the objective survival rates $s(t, H)$; only the agent's Bellman equation uses s^{subj}. A per-year factor of $\psi = 0.981$ matches the O'Dea–Sturrock finding: over a 10-year horizon,

$(0.71/0.86)^{1/10} \approx 0.981$. When $\psi = 1$, beliefs are objective and this channel is inactive.

2.5 Medical expenditures

Out-of-pocket medical expenditures depend on age and health. Log medical spending follows:

$$\ln m_t = \mu_m(t, H_t) + \sigma_m(t, H_t) \cdot \varepsilon_t, \quad \varepsilon_t \sim N(0, 1), \tag{6}$$

where μ_m and σ_m are the mean and standard deviation of log expenditures, both age- and health-dependent. The baseline calibration matches the age profiles reported by Jones et al. (2018): mean out-of-pocket spending of \$4,200 at age 70, rising to \$29,700 at age 100, with a 95th percentile of approximately \$111,200 at age 100.

A Medicaid safety net provides a consumption floor $\underline{c}$. If total resources (wealth plus income minus medical costs) fall below $\underline{c}$, the government covers the shortfall. The individual consumes $\underline{c}$ and enters the next period with zero liquid wealth. This floor is calibrated to the SSI federal benefit rate plus average state supplement for an elderly individual (\$6,180, following the value used in Lockwood, 2012).

Expectations over medical expenditure shocks are computed using nine-node Gauss–Hermite quadrature. At the calibrated variance ($\sigma \approx 1.4$), lower-order rules place insufficient weight in the tails of the lognormal distribution; Appendix G documents the convergence diagnostics.

2.6 Annuity pricing

The annuity is nominal: the insurer prices the stream of nominal payments using the nominal discount rate $r_{\text{nom}} = (1+r)(1+\pi) - 1$, where r is the real interest rate and π is the inflation rate. The actuarially fair annual payout per dollar of premium is:

$$\text{payout}_{\text{fair}} = \frac{1}{\sum_{k=0}^{T-1} \frac{\prod_{j=0}^{k-1} s_{\text{base}}(j)}{(1+r_{\text{nom}})^k}}. \tag{7}$$

The nominal discount rate reflects the insurer's investment in nominal bonds. The resulting initial payout is higher than under real discounting, but the consumer's real income from the annuity declines each period:

$$A_t^{\text{real}} = \frac{A_1}{(1+\pi)^{t-1}}, \tag{8}$$

where π is the annual inflation rate. At $\pi = 2\%$, purchasing power falls by 39% over 25 years.

When $\pi = 0$ (the Yaari benchmark and intermediate decomposition steps where inflation is inactive), $r_{\text{nom}} = r$ and the pricing reduces to the standard real-annuity formula.

The annuity is priced with a money's worth ratio (MWR), defined as the expected present value of payouts per dollar of premium. The loaded payout rate is MWR $\times$ $\text{payout}_{\text{fair}}$. I set MWR = 0.87 at baseline, consistent with the population-table estimates of Mitchell et al. (1999) for the US individual annuity market. This choice merits discussion. Mitchell et al. (1999) estimated MWRs of 0.80–0.85 using population mortality tables, which is the relevant benchmark for an individual considering annuitization (the "money's worth" of the purchase against population life expectancy). Using annuitant mortality tables—which adjust for selection—yields higher MWRs of 0.90–0.94, but this overstates the value proposition for the marginal non-annuitant. More recent estimates from Wettstein et al. (2021) suggest some improvement, with population-table MWRs around 0.85. Section 5 reports full sensitivity: at MWR = 0.85, predicted ownership rises to 12.5%; at MWR = 0.90, to 26.2%. The steep sensitivity to MWR is itself a finding—it identifies pricing as the most actionable policy lever.

A fixed cost of $1,000 is incurred upon any purchase.[2]

[2]The MWR is computed using beginning-of-period payment timing: the first payment occurs at purchase. This convention is standard in the annuity pricing literature (Mitchell et al., 1999) and matches the implementation in the replication code.

2.7 Budget constraint

Within each period, the timing is:

1. The individual enters with liquid wealth W_t, annuity income A_t, Social Security $\mathrm{SS}(t)$, and health H_t.
2. Medical expenditure m_t is realized.
3. The individual chooses consumption c_t.
4. Remaining wealth earns the risk-free return r.
5. Health transitions to H_{t+1}.
6. The individual survives to $t+1$ with probability $s(t, H_t)$.

The intertemporal budget constraint is:

$$W_{t+1} = (1+r)(W_t + A_t + \mathrm{SS}(t) - m_t - c_t), \tag{9}$$

subject to $c_t \geq \underline{c}$ and $W_{t+1} \geq 0$ (no borrowing).

2.8 Age-varying consumption needs

Aguiar and Hurst (2013) document that consumption expenditure declines at roughly 2% per year in retirement, even among wealthy households, once time costs and home production are accounted for. This decline reflects changing consumption needs—reduced transportation, clothing, and food-away-from-home expenditures—rather than binding liquidity constraints. The declining profile reduces the value of late-life income streams, including annuities.

The model captures this through an age-dependent weight on flow utility:

$$w(t) = (1-\delta_c)^{t-1}, \quad \delta_c = 0.02, \tag{10}$$

where $t = 1$ at age 65. Flow utility becomes $w(t) \cdot u(c)$. This specification is equivalent to an age-dependent felicity shifter: as consumption needs decline, the marginal value of additional consumption in late life falls, reducing the insurance value of an annuity that provides a level real income stream. When $\delta_c = 0$, this channel is inactive and the model reduces to the standard time-separable case.

Age-varying consumption needs should be distinguished from time preference (β). The discount factor β governs the rate at which future utility is discounted; $w(t)$ governs how much utility a given level of consumption generates at each age. A constant β with declining $w(t)$ implies that the individual values future consumption opportunities but needs less consumption to achieve the same felicity as she ages.

2.9 State-dependent utility

Finkelstein et al. (2013) estimate that the marginal utility of consumption declines with poor health, with a central estimate implying that individuals in poor health value a dollar of consumption roughly 10–25% less than those in good health. Reichling and Smetters (2015) incorporated this channel in their annuity model. State-dependent utility reduces annuity demand because the annuity delivers income in poor-health states where its marginal utility is lower.

The model allows the marginal utility of consumption to depend on health status through a multiplicative weight:

$$u(c, H) = \varphi(H) \cdot \frac{c^{1-\gamma}}{1-\gamma}, \tag{11}$$

where $\varphi(\text{Good}) = 1.00$, $\varphi(\text{Fair}) = 0.90$, and $\varphi(\text{Poor}) = 0.75$, following the mapping used by Reichling and Smetters (2015) based on the estimates of Finkelstein et al. (2013). When $\varphi(H) = 1$ for all H, this channel is inactive.

In practice, state-dependent utility has a negligible effect on predicted ownership (Shapley value of 1.1 pp). The channel is included for completeness and to confirm that its quantitative

irrelevance, suggested by the prior literature, holds in the unified framework.

2.10 Two behavioral channels

The nine rational and preference-based channels omit two well-documented forces that operate at distinct decision moments and point in opposite directions. The first concerns how households consume out of different sources of wealth *after* a purchase has been made; the second concerns the framing of the irreversible purchase *at* the decision moment. Both are needed because the empirical evidence supports each independently, and because suppressing either to make the model look "tame" would discard the data (Thaler, 1999; Barberis and Huang, 2009).

Force A: Source-dependent utility. Blanchett and Finke (2024, 2025) document a robust spending differential between retirees consuming guaranteed income (Social Security, annuity payouts) and those drawing from portfolio assets: retirees spend roughly 80% of guaranteed income but only 50% of common withdrawal benchmarks (e.g., the 4% rule) from portfolio wealth. The 50/80 ratio implies a "license to spend" embedded in income flows that is absent from portfolio drawdowns, consistent with the mental-accounting tradition initiated by Shefrin and Thaler (1988) and Thaler (1999).

The model captures this through an effective-consumption transformation. Within each period, total consumption c is decomposed into income- and portfolio-financed components:

$$c_{\text{income}} = \min(c, \text{SS} + A), \qquad c_{\text{portfolio}} = \max(0, c - \text{SS} - A). \tag{12}$$

Effective consumption is then

$$c_{\text{eff}} = c_{\text{income}} + \lambda_W \cdot c_{\text{portfolio}}, \qquad \lambda_W \in (0, 1], \tag{13}$$

and per-period utility is $u(c_{\text{eff}}, H)$. The discount is applied at the dollar level (not the utility

level) to avoid the sign-flip that would occur for $\gamma > 1$ if λ_W multiplied a negative CRRA value directly. Calibration: $\lambda_W = 0.625 = 50/80$, anchored to the Blanchett–Finke spending differential. This is identical to the source-dependence parameter in a companion paper on Social Security claiming (Tharp, 2026, under review); cross-paper consistency converts one of two free parameters into an out-of-sample prediction. When $\lambda_W = 1$ this channel is inactive.

Force B: Narrow-framing purchase penalty. Barberis and Huang (2009) document that consumers evaluate gambles in narrow brackets rather than integrating them with total wealth, applying loss aversion to the running gain/loss tally on each isolated investment. Chalmers and Reuter (2012) find that voluntary annuitization in Oregon PERS administrative data was approximately 50% under default and 15% under opt-in, a 35 percentage point active-purchase wedge that pricing cannot explain. The UK 2015 pension freedoms reform changed the default architecture from compulsory annuitization to opt-in; annuity sales fell by approximately 75–87% (Association of British Insurers, 2020). Goda et al. (2014) document an analogous gap in the federal Thrift Savings Plan annuity option (≈86 pp). Brown et al. (2008) find that consumption-frame versus investment-frame elicitations of annuity demand differ by 50 percentage points.

In our model, the household evaluates the SPIA as a stand-alone investment with its own gain/loss tally—the cumulative annuity payout minus the premium paid. While this tally is negative (cumulative payouts < premium), the household is "underwater" and experiences a per-period loss-aversion flow proportional to the unrecouped premium. Once cumulative payouts cross the premium (*breakeven*, approximately age 79 for typical SPIA pricing), the loss tally turns positive and the penalty vanishes. The per-period flow at period t (where $t = 1$ corresponds to age 65 and A is the annual annuity income) is

$$\text{flow}_t = \psi_{\text{purchase}} \cdot u'(c_{\text{ref}}) \cdot \max\Big(0,\ \text{premium} - A \cdot (t-1)\Big), \tag{14}$$

where $u'(c_{\text{ref}}) = c_{\text{ref}}^{-\gamma}$ is the marginal utility at the reference consumption level (we use $c_{\text{ref}} =$ \$18,000, the population mean Social Security benefit). The total disutility entering the age-65 alpha search is the survival- and discount-weighted NPV of this stream:

$$\text{Penalty}_{\text{NPV}} = \sum_{t=1}^{t^*} \beta^{t-1} \cdot S(t) \cdot \text{flow}_t, \tag{15}$$

where $S(1) = 1$ and $S(t) = \prod_{s=1}^{t-1} s_s$ is the cumulative survival probability, and t^* is the breakeven horizon. This formulation is derivable from Barberis and Huang (2009) narrow framing combined with Tversky and Kahneman (1992) loss aversion: the household experiences amplified disutility from the negative side of the running gain/loss tally each period until it turns positive.

Calibration of ψ_{purchase}. We calibrate ψ_{purchase} via single-moment SMM to a behavioral elasticity from the UK 2015 pension freedoms reform, an external natural experiment. The reform flipped the choice architecture from compulsory annuitization to opt-in; UK annuity sales fell roughly 75–87 percentage points (Association of British Insurers, 2020). Approximately 15–25 pp of this drop reflects a rational response to the simultaneous removal of a 55% tax penalty on lump-sum withdrawals; the remaining 55–65 pp is the behavioral residual attributable to architecture change alone. We target the midpoint of this behavioral range (60 pp) and find the value of ψ_{purchase} that matches the model's no-PED-to-full-PED ownership drop. The result is $\psi_{\text{purchase}} = 0.0163$, with sensitivity across alternative single-anchor SMM specifications of $\psi_{\text{purchase}} \in [0.014, 0.028]$ (Appendix A.7).

Why the behavioral-residual anchor. Three considerations drive the choice of the rational-corrected behavioral residual as the primary calibration target. First, ψ_{purchase} in the structural model is a preference primitive (loss aversion over a narrowly framed gain/loss tally); the 55% tax penalty operates through the budget constraint and is already represented in the model's rational pricing channels. Calibrating ψ_{purchase} to the raw 75–87 pp drop would

force the parameter to absorb a wedge the structural model handles elsewhere—a textbook over-identification failure. Second, under Heckman and Pinto (2018) transportability, structural preference parameters transport across environments via elasticities (architectural-default response), not levels. UK and US differ in tax structure, State Pension generosity, bequest culture, and wealth distribution; matching levels (UK 17% retention) would require these channels to wash out identically across countries, which is implausible. Third, the rational-corrected behavioral elasticity preserves interior identification: at $\psi_{\text{purchase}} = 0.0163$ the model produces a non-corner ownership prediction, while higher anchors (matching the raw 78 pp UK drop) push the model to a corner where any sufficiently large ψ_{purchase} produces an identical 0% prediction.

Out-of-sample test. The calibrated ψ_{purchase} is set entirely from UK data, with no reference to observed US ownership. The resulting US predicted voluntary annuitization rate at the production calibration is therefore an out-of-sample prediction. Section 4 reports this prediction against the observed 3.4% rate in the HRS sample.

Non-overlap. The two forces target empirically distinct margins. Source-dependent utility operates on the *ongoing consumption* from existing wealth (post-purchase, every period); it shows up in passive spending patterns documented by Blanchett–Finke and in the fungibility violations documented by Hastings and Shapiro (2013). The narrow-framing penalty operates on the *at-purchase decision* (one-time, with stream-form ongoing pain until breakeven); it shows up at transaction boundaries documented by Brown et al., the UK 2015 quasi-experiment, and the TSP defaults. Section 4 reports the joint Shapley decomposition; the empirical signatures are distinct.

2.11 Bellman equation

For $t < T$, the value function satisfies:

$$V(W, A, H, t) = \max_c \Big\{ w(t) \cdot u(c, H) \\ + \beta\, s(t, H)\, \mathbb{E}\Big[V(W', A', H', t+1) \mid H\Big] \\ + \beta\,(1 - s(t, H))\, v(W') \Big\}, \tag{16}$$

where $W' = (1 + r)(W + A_t + \mathrm{SS}(t) - m - c)$, $A' = A/(1 + \pi)$ is next period's real annuity income, and the expectation is taken over the joint distribution of H' (health transition) and m (medical shock). The term $w(t)$ is the age-varying needs weight from equation (10) and $u(c, H)$ is the state-dependent utility from equation (11). When both channels are inactive ($\delta_c = 0$, $\varphi(H) = 1$), this reduces to the standard Bellman equation. At the terminal period T, the individual consumes all remaining resources and leaves any residual as a bequest.

At $t = 1$ (age 65), the annuitization decision is:

$$\alpha^* = \arg\max_{\alpha \in [0,1]} V\Big((1 - \alpha)W_0 - F \cdot \mathbf{1}(\alpha > 0),\ \alpha W_0 \cdot \text{payout},\ H_0,\ 1\Big), \tag{17}$$

where F is the fixed purchase cost.

2.12 Analytical characterization

The numerical decomposition in Section 4 can be given analytical structure. Consider the annuitization decision at $\alpha = 0$. Shifting one dollar from liquid wealth to the annuity premium changes welfare by

$$\left.\frac{dV}{d\alpha}\right|_{\alpha=0} = W_0\Big[\mathrm{MWR} \cdot V_A(W_0, 0, H, 1) - V_W(W_0, 0, H, 1)\Big], \tag{18}$$

where V_W and V_A are the partial derivatives of the value function with respect to liquid wealth and annuity income. Define the *annuity value ratio*

$$R \equiv \text{MWR} \cdot \frac{V_A}{V_W}. \tag{19}$$

The agent annuitizes a positive fraction if and only if $R > 1$.

Proposition 1 (Yaari)**.** *With no bequests ($\theta = 0$), deterministic health, fair annuity pricing, and $\beta(1+r) = 1$, the envelope theorem implies $V_W = u'(c^*)$ at the optimum. The annuity-income derivative V_A is the present discounted value of future marginal utilities generated by the income stream, weighted by survival probabilities. Consumption smoothing across states equates these marginal values, yielding $R = 1$ with full annuitization weakly optimal.*

Each demand-suppressing channel introduces a factor $\Phi_i < 1$ that reduces R below the Yaari benchmark:

Proposition 2 (Super-additive decomposition)**.** *The annuity value ratio can be written*

$$R = \Phi_{MWR} \times \Phi_{bequest} \times \Phi_{health} \times \Phi_{inflation} \times \Phi_{SS} \times \Phi_{pessimism} \times \Phi_{age} \times \Phi_{state}, \tag{20}$$

where:

- $\Phi_{MWR} = MWR$*. The pricing load directly scales the numerator. At $MWR = 0.87$, each dollar of premium buys only 82 cents of expected value.*
- $\Phi_{bequest} = \left[1 + (1-s)\, v'(b)/u'(c)\right]^{-1}$*, where s is the survival probability and $v'(b)/u'(c)$ is the ratio of marginal bequest to marginal consumption utility. Under the DFJ luxury-good specification ($\kappa = \$272{,}628 \gg$ median wealth $\approx \$40{,}000$), $v'(b) \approx \theta(\kappa)^{-\gamma}$ is nearly constant and small, so $\Phi_{bequest} \approx 1.00$.*
- Φ_{health} *reflects the covariance between survival and the marginal utility of liquid wealth. When health deterioration simultaneously reduces survival and raises medical costs, the*

ratio V_A/V_W falls. This is the Reichling–Smetters mechanism.

- $\Phi_{inflation}$ *equals the ratio of the present value of the nominal payment stream (in real terms) to the real payment stream, weighted by marginal utilities. At 2% inflation, payments in year 25 retain only 61% of their initial purchasing power.*

- Φ_{SS} *captures the diminishing marginal insurance value of additional annuitization when Social Security already provides a longevity-insured income floor.*

- $\Phi_{pessimism} < 1$ *because the agent uses $\psi \cdot s(t, H)$ in the Bellman equation. The subjective present value of future annuity income is lower than the objective present value used by the insurer to set the price.*

- $\Phi_{age} < 1$ *because declining consumption needs $w(t)$ reduce the marginal utility of late-life income, which is precisely the income the annuity provides.*

- $\Phi_{state} \leq 1$ *because poor-health states generate lower marginal utility of consumption, and the annuity pays into those states.*

The two behavioral channels do not enter equation (20) as smooth multiplicative factors on R. Source-dependent utility shifts V_A upward (the annuity stream is consumed under full utility weight while portfolio drawdowns are discounted at λ_W), so the SDU effect is best read as a multiplicative factor $\Phi_{\text{SDU}} > 1$ on the numerator of R. The narrow-framing purchase penalty (Force B) does not fit the marginal-margin framework at all: it is a fixed disutility subtracted from the value function at the age-65 alpha search (equation 15), shifting the participation threshold rather than the marginal value ratio. The Shapley decomposition in Section 4.4, which is computed on the full discrete-choice participation outcome, captures both forces directly.

Proposition 3 (Super-additivity of participation effects)**.** *The fraction of agents annuitizing is* $\Pr(\alpha^* > 0) = \Pr(R_i > 1)$*, where R_i varies across agents due to heterogeneity in wealth,*

health, and Social Security income. In log space,

$$\ln R_i = \sum_j \ln \Phi_j + \varepsilon_i, \tag{21}$$

where ε_i captures agent-level heterogeneity. Each additional channel shifts the distribution of $\ln R_i$ leftward by $|\ln \Phi_j|$. Agents near the participation threshold ($R_i \approx 1$, i.e., $\ln R_i \approx 0$) are disproportionately eliminated, so the ownership drop from adding a channel is larger when other channels have already thinned the right tail of the R_i distribution. This super-additive structure motivates the exact Shapley decomposition in Section 4.4, which attributes the total demand reduction to individual channels without dependence on the order in which channels are added.

The dominant term is $\Phi_{\text{MWR}} = 0.87$, which alone reduces the annuity value ratio by 13%. The near-unity Φ_{bequest} under the DFJ specification explains why bequests contribute only 1.0 pp in the Shapley decomposition: the luxury-good curvature parameter makes bequest utility irrelevant for median-wealth retirees.

2.13 Solution method

The model is solved by backward induction over the state space (W, A, H, t). The wealth grid uses 80 points with power-function spacing (denser at low wealth) over $[0, \$3 \text{ million}]$, covering the 99.5th percentile of the HRS wealth distribution. The annuity income grid uses 30 points with cubic spacing to resolve small purchases, and health is discrete with three states. Consumption is optimized at each grid point using Brent's method. Value function interpolation across the wealth dimension uses piecewise linear interpolation with flat extrapolation.

The annuitization fraction α is optimized over a 101-point grid at age 65. Grid convergence is verified: ownership varies by only 1.1 percentage points across grid resolutions from 60×20 to 120×50 at the 9-node production quadrature rule (see Appendix Table 4).

3 Calibration and Validation

Table 1 summarizes the parameter values. Each is drawn from a published source or calibrated to match moments from the HRS.

Preferences. I set $\gamma = 2.5$, within the range of 1–3 estimated by Chetty (2006) from labor supply elasticities. Lockwood (2012) uses $\gamma = 2$; the slightly higher value reflects the richer model environment. Section 5 reports results for $\gamma \in \{1.5, 2.0, \ldots, 5.0\}$.

Age-varying consumption needs. The decline rate $\delta_c = 0.02$ matches the central estimate from Aguiar and Hurst (2013), who found that nondurable consumption expenditure declines at roughly 2% per year in retirement after controlling for changes in household composition. This rate is applied uniformly across health states.

State-dependent utility. The health-utility weights $\varphi = [1.00, 0.90, 0.75]$ follow the mapping used by Reichling and Smetters (2015), based on the estimates of Finkelstein et al. (2013) from the HRS. Good-health individuals receive full weight; Fair and Poor health reduce the marginal utility of consumption proportionally.

Source-dependent utility (Force A). The production calibration sets $\lambda_W = 0.625 = 50/80$, anchored to the empirical spending differential documented by Blanchett and Finke (2024, 2025): retirees consume approximately 80% of guaranteed income but only 50% of common withdrawal benchmarks (e.g., the 4% rule) from portfolio wealth. The 50/80 ratio is the implied utility weight on portfolio-financed consumption relative to income-financed consumption. This is identical to the source-dependence parameter in a companion paper on Social Security claiming (Tharp, 2026, under review), providing cross-paper consistency: λ_W is fixed before this paper's predictions are computed.

Table 1: Model Parameters

Parameter	Symbol	Value	Source
Preferences			
Risk aversion	γ	2.5	Within Chetty (2006) range (1–3)
Discount factor	β	0.97	Standard
Bequest intensity	θ	56.96	Lockwood (2012)
Bequest shifter	κ	\$272,628	Lockwood (2012); De Nardi (2004)
Demographics			
Entry age		65	Retirement
Maximum age		110	Lockwood (2012)
Risk-free rate	r	0.02	Real
Health and mortality			
Health states		3	Good, Fair, Poor
Hazard multipliers	$\mu(H,t)$	See text	HRS mortality data (see text)
Quadrature nodes		9	Gauss–Hermite
Survival pessimism	ψ	0.981	O'Dea and Sturrock (2023)
Medical expenditures			
Log mean at 65 (Fair)	μ_m	7.037	Jones et al. (2018)
Annual growth		0.065	Jones et al. (2018)
Log std dev	σ_m	1.4	Jones et al. (2018)
Consumption floor	$\underline{c}$	\$6,180	SSI + state supplement
Annuity market			
Money's worth ratio	MWR	0.87	Mitchell et al. (1999)
Fixed cost	F	\$1,000	Lockwood (2012)
Inflation rate	π	0.02	Post-Volcker average
Preference and behavioral channels			
Age-varying needs decline	δ_c	0.02	Aguiar and Hurst (2013)
Health-utility weights	$\varphi(H)$	$[1.00, 0.90, 0.75]$	Finkelstein et al. (2013)
Source-dependent utility (Force A)	λ_W	0.625	Blanchett and Finke (2024, 2025); Shefrin and Thaler (1988); Thaler (1999)
Narrow-framing penalty (Force B)	ψ_{purchase}	0.0163	UK 2015 reform (Association of British Insurers, 2020); SMM

All dollar values in 2014 dollars.

Narrow-framing purchase penalty (Force B). The production calibration sets $\psi_{\text{purchase}} = 0.0163$ via single-moment simulated method of moments (SMM) on the rational-corrected behavioral elasticity from the UK 2015 pension freedoms reform. The reform flipped the choice architecture from compulsory annuitization to opt-in, producing a 75–87 percentage point drop in annuity sales (Association of British Insurers, 2020). Approximately 15–25 pp of the drop reflects the rational response to simultaneous removal of a 55% tax penalty on lump-sum withdrawals; the remaining 55–65 pp is the behavioral residual. We target the midpoint of this behavioral range (60 pp) and bisect on ψ_{purchase} to match. Across alternative single-anchor SMM specifications—matching UK retention level, raw total elasticity, or the high/low ends of the behavioral residual—the implied ψ_{purchase} falls in $[0.0142, 0.0281]$ and the resulting predicted US ownership ranges from 2.2% to 22.0% (Appendix A.7). The production point estimate sits at the midpoint of this interval. A key feature of this calibration approach: ψ_{purchase} is identified from a non-US natural experiment that perturbs the choice architecture while holding prices, products, and preferences fixed. The resulting US ownership prediction is therefore an out-of-sample test, not a calibration target.

The bequest parameters $\theta = 56.96$ and $\kappa = \$272{,}628$ are taken directly from Lockwood (2012), who estimated them from HRS bequest data at $\gamma = 2$. Using them at $\gamma = 2.5$ raises a portability concern in principle, but simulation-based checks show less than 14% divergence in the bequest-to-wealth ratio across $\gamma \in [1.5, 5.0]$, below the threshold warranting recalibration. All results use Lockwood's original estimates.

Health and mortality. Health transition matrices are calibrated to HRS panel data following De Nardi et al. (2010). The five HRS self-reported health categories are mapped to three states: Good (Excellent/Very Good), Fair (Good), and Poor (Fair/Poor). Baseline survival probabilities use the SSA administrative life table from Lockwood (2012).

Hazard multipliers are estimated from the RAND HRS longitudinal file (waves 4–16, $N = 126{,}249$ person-wave observations aged 65+). The estimated ratios by age band are:

Age band	Good	Fair	Poor
65–74	0.49	1.00	3.29
75–84	0.60	1.00	2.77
85+	0.74	1.00	1.82

The compression of the gradient at older ages is consistent with selection effects: those surviving to 85 in Poor health have already demonstrated above-average resilience. The baseline constant multipliers $[0.50, 1.0, 3.0]$ represent a central tendency across these age bands, falling between the R-S functional estimates ($[0.45, 1.0, 3.5]$, consistent with ADL/IADL-based measures) and the empirical HRS self-reported health values ($[0.57, 1.0, 2.7]$). Section 5 reports results across the full range.

Medical expenditures. The medical expenditure process matches the moments published by Jones et al. (2018): mean out-of-pocket spending of $4,200 at age 70, growing to $29,700 at age 100, with a 95th percentile of approximately $111,200 at age 100. Health shifts both the mean and variance of log spending, with Poor health raising expected costs by a factor of roughly two relative to Fair.

Population sample. The wealth distribution is drawn from the RAND HRS longitudinal file, selecting single retirees aged 65–75 using filters comparable to Lockwood (2012). The resulting sample contains 793 individuals. The main analysis restricts to the 566 individuals with liquid wealth above $5,000, as those with negligible wealth cannot meaningfully annuitize. Full-sample results (including zero-wealth individuals) are reported for comparison with observed ownership rates of 3–6%. In the pooled HRS sample (5,303 person-wave observations, waves 5–9), 3.4% report positive annuity income, consistent with Lockwood (2012)'s 3.6% estimate. The analysis uses unweighted observations; the HRS sample is approximately representative of the target population (single retirees aged 65–75), and the structural model maps each household's state to a predicted decision rather than estimating moments from

the sample distribution.

Lifecycle moment validation. Table 2 compares simulated lifecycle moments against empirical targets across four dimensions: bequest distribution, medical spending, survival, and health prevalence. None of these moments are targeted in calibration; the comparison serves as an external check on the model's internal consistency.

Bequests match well at the median, within roughly 5% of the HRS estimate, though the simulated mean understates the empirical target by reflecting a thinner right tail in the model. Medical spending matches the Jones et al. (2018) target at age 70 closely but overstates at older ages, partly because health composition in the simulated population (which starts in Fair health) is worse than in the data. Survival at age 75 matches the SSA life table closely; modest divergence at age 85 is consistent with the faster health deterioration path in the simulation. Health prevalence at age 75 shows the expected artifact of the conditional simulation design: starting all agents in Fair health understates Good prevalence and overstates Poor. Table 2 reports the full age-by-age comparison.

Table 2: Simulated vs Empirical Lifecycle Moments

Moment	Simulated	Empirical (HRS)
Mean bequest	\$28946	\$90000
Median bequest	\$0	\$20000
Fraction bequest > \$10K	26.4%	45.0%

Simulated: 100,000 trajectories, initial wealth \$250,000, Fair health.
Empirical: HRS exit interviews (bequests); Jones et al. (2018) (medical).

4 Results

4.1 Sequential decomposition

Table 3 presents an intuitive sequential decomposition of predicted voluntary annuity ownership, adding the seven standard rational channels one at a time. Because this exercise

is order-dependent, I use it as a descriptive path through the model and treat the exact Shapley values below as the preferred attribution.

Table 3: Sequential Decomposition of Predicted Voluntary Annuity Ownership

Channel	Ownership (%)	Δ (pp)	Retention	Cumulative
Yaari benchmark	41.4	—	—	—
+ Social Security	100.0	+58.6	241.7%	2.4170
+ Bequest motives	100.0	+0.0	100.0%	2.4170
+ Medical expenditure risk (uncorrelated)	97.4	-2.6	97.4%	2.3533
+ Health-mortality correlation (R-S)	93.3	-4.0	95.8%	2.2556
+ Survival pessimism	80.2	-13.1	86.0%	1.9388
+ State-dependent utility	81.0	+0.7	100.9%	1.9567
+ Age-varying consumption needs	69.3	-11.7	85.6%	1.6747
+ Realistic pricing loads	34.7	-34.6	50.1%	0.8390
+ Inflation erosion	22.3	-12.4	64.2%	0.5384
+ Source-dependent utility	77.2	+54.9	346.4%	1.8649
+ Purchase disutility	17.4	-59.7	22.6%	0.4212
Observed (Lockwood 2012)	3.6	—	—	—

Retention rate = ownership after channel / ownership before channel. Cumulative product of retention rates tracks geometric compounding.

Frictionless population benchmark. The first row—no bequest motive, actuarially fair pricing, no medical expenditure risk, no inflation, and no Social Security income—predicts 41.4% ownership in the HRS population sample. This is not the theoretical Yaari full-annuitization result; it is the frictionless participation benchmark implied by the empirical wealth distribution and a positive fixed purchase cost. Many low-wealth households in the data still do not find annuitization worthwhile even before the main demand-suppressing channels are activated.

Social Security. Adding Social Security income (COLA-protected, calibrated by wealth quartile) raises predicted ownership to 100.0%. SS acts as a complement in the frictionless model: the guaranteed income floor enables consumption smoothing and makes agents willing to lock up additional wealth in annuities. The 241.7% retention rate reflects this

complementarity.

Bequest motives. Adding the DFJ luxury-good bequest specification ($\theta = 56.96$, $\kappa =$ \$272,628) leaves ownership unchanged at 100.0% (retention rate of 100%). The luxury-good $\kappa =$ \$272,628 makes marginal bequest utility nearly flat for individuals with wealth below \$200,000. In the HRS sample, median liquid wealth is approximately \$40,000.

Medical expenditure risk. Introducing stochastic medical expenditures—without linking them to mortality—reduces ownership to 97.4% (retention 97.4%, a −2.6 pp effect). At this stage, survival probabilities are identical across health states, so the annuity remains a pure longevity hedge.

Health-mortality correlation. Activating the Reichling–Smetters mechanism reduces ownership to 93.3% (retention 95.8%). When a retiree's health deteriorates, expected remaining lifetime falls (reducing the present value of future annuity payments) and expected medical costs rise (increasing the marginal value of liquid wealth). The distinction between medical risk and health-mortality correlation is economically substantive: the demand-suppressing effect depends specifically on the correlation between health deterioration, survival reduction, and cost increases. This channel was not jointly incorporated in the multi-channel models of Pashchenko (2013) or Peijnenburg et al. (2016).

Survival pessimism. Introducing subjective survival beliefs ($\psi = 0.981$) reduces ownership from 93.3% to 80.2% (retention 86.0%, a −13.1 pp effect). The agent overweights the probability of dying before recovering the premium.

Pricing loads. Repricing the annuity at MWR = 0.87 and adding a \$1,000 fixed purchase cost produces the largest single reduction, from 80.2% to 44.3% (retention 55.2%). An 13% load eliminates the surplus that marginally willing buyers retained after accounting for health risk and survival pessimism.

Inflation erosion. Converting the annuity from real to nominal at 2% annual inflation reduces ownership from 44.3% to 35.6% (retention 80.3%). This step is a product counterfactual: the previous steps priced a real annuity (constant purchasing power); this step switches to a nominal annuity whose real value erodes over time. Social Security income is COLA-protected and does not erode, but the private annuity payment loses purchasing power: at 2% inflation, real income from the annuity falls by 39% over 25 years.

Summary. The seven standard rational channels reduce predicted ownership from 41.4% to 35.6%, a cumulative retention product of 0.443. This 35.6% prediction is close to the approximately 20% participation predicted by Pashchenko (2013) using a different but overlapping channel set. Pricing loads (retention 55.2%) and inflation erosion (retention 80.3%) are the two most powerful channels in this sequential ordering. The key implication is that the standard rational literature explains much of the gap, but not all of it.

4.2 Preference Channel Extensions

Table 4 isolates the incremental effect of the two added preference channels.

Table 4: Four-Layer Decomposition: Rational, Preference, and Two Behavioral Channels

Specification	Ownership (%)	Δ (pp)
Seven rational channels (Layer 1)	35.6	—
+ Age-varying consumption needs	22.7	-12.8
+ State-dependent utility (Layer 2 complete)	22.3	-0.4
+ Source-dependent utility (Force A)	77.2	+54.9
+ Narrow-framing penalty (Force B)	17.4	-59.7

Layer 1 is the standard rational decomposition. Layer 2 adds the two preference channels (age-varying needs from Aguiar and Hurst, 2013; state-dependent utility from Finkelstein et al., 2013). Force A introduces source-dependent utility (Blanchett and Finke, 2024, 2025): portfolio-financed consumption is discounted relative to income-financed consumption, mirroring the specification in the FPR companion paper. Force B introduces a narrow-framing purchase penalty (Barberis and Huang, 2009; Tversky and Kahneman, 1992): a per-period loss-aversion flow over the unrecouped premium, decaying to zero at breakeven.

Adding age-varying consumption needs ($\delta_c = 0.02$, calibrated to Aguiar and Hurst, 2013) reduces predicted ownership from 35.6% to 22.7%. This is the quantitatively important extension beyond the seven-channel rational model: if late-life consumption needs decline, the insurance value of a level annuity stream declines as well. The resulting reduction is comparable in magnitude to survival pessimism or inflation erosion.

Adding state-dependent utility ($\varphi = [1.00, 0.90, 0.75]$, taken directly from the central estimates of Finkelstein et al., 2013) produces only a small additional reduction, from 22.7% to 22.3%. The channel's effect is insensitive to the specific mapping within the range the literature supports: under the softer translation $\varphi = [1.00, 0.95, 0.85]$ used by Reichling and Smetters (2015), the full-model prediction shifts to 22.7%, a difference of 0.4 pp. In either calibration, this channel is best interpreted as a completeness check rather than a load-bearing mechanism.

The full nine-channel model therefore predicts 22.3% ownership under the raw FLN central mapping and 22.7% under the Reichling–Smetters translation, a substantial reduction from the 41.4% frictionless benchmark but still above the 3.4% observed in the present sample and the 3–6% range reported by Lockwood (2012) and Dushi and Webb (2004). The remainder of the paper uses the raw FLN mapping as the production calibration.

4.3 Behavioral channel extensions: the two-force structure

The nine-channel model (rational stack plus the two preference channels) predicts 22.3% ownership, well above the 3.4% observed in the present sample and the 3–6% reported by Lockwood (2012) and Dushi and Webb (2004). The two behavioral channels operate at distinct decision moments and produce a countervailing pattern in the sequential decomposition.

Activating the source-dependent utility channel (Force A, $\lambda_W = 0.625$) raises predicted ownership from 22.3% to 77.2%—a roughly +55 pp boost. The mechanism is the "license to spend" embedded in income flows: once a household annuitizes, the resulting income stream

is consumed under the full utility weight ($\lambda_W = 1$ for income), while remaining portfolio wealth is consumed under the discounted weight ($\lambda_W = 0.625$). Annuitization therefore expands the share of consumption that is psychologically frictionless, raising the ex-ante value of the annuity at every margin where the household would otherwise have drawn from portfolio.

Activating the narrow-framing purchase penalty (Force B, $\psi_{\text{purchase}} = 0.0163$) on top of source-dependent utility produces the full eleven-channel prediction of 17.4%—a roughly −60 pp suppression relative to the SDU-only intermediate. The mechanism is loss aversion over the cumulative gain/loss tally on the annuity "investment": until cumulative payouts cross the premium (breakeven, approximately age 79), the household is underwater and experiences per-period loss-aversion disutility. The penalty's NPV at the age-65 alpha search is large enough to reverse the SDU boost for most marginal annuitizers but not all.

The two forces are individually large in absolute magnitude (Force A roughly +55 pp, Force B roughly −60 pp) and partly offset; the net behavioral effect on the headline (sequential drop from 22.3% to 17.4%, about −5 pp) is much smaller than either force in isolation. The full eleven-channel model is therefore the paper's production specification. Both behavioral parameters are identified entirely out of sample. λ_W matches the Blanchett–Finke 80%/50% spending differential, set ex ante in a companion paper. ψ_{purchase} is calibrated by single-moment SMM to the rational-corrected behavioral elasticity from the UK 2015 pension freedoms reform (60 pp drop, midpoint of the 55–65 pp behavioral range; see Section 2.10).

Out-of-sample test. The calibrated ψ_{purchase} uses no US ownership information; the predicted US voluntary annuitization rate is therefore an out-of-sample test against the observed HRS rate of 3.4%. Across alternative single-moment SMM specifications, the predicted ownership ranges from 2.2% to 22.0% (Appendix A.7), a bracket that contains observed 3.4%. The midpoint anchor (Anchor C-mid: 60 pp behavioral residual after correcting for the simultaneous tax-rule change in the UK reform) gives the production point estimate of 17.4%,

an overshoot relative to observed of roughly 14 pp. This residual is informative about US-specific factors not in the UK calibration—advisor incentives, distribution channel frictions, and product complexity that the structural model does not separately represent—rather than evidence against the model's structural account. The ψ_{purchase} value that would match observed ownership exactly sits within the bracket at roughly the upper end of the behavioral residual range.

4.4 Exact Shapley decomposition

The sequential decomposition is order-dependent: the contribution attributed to each channel depends on which channels precede it. To obtain order-independent attributions, I compute exact Shapley values by solving the model for all $2^{11} = 2{,}048$ channel subsets and averaging each channel's marginal contribution across all possible orderings.

Table 5 reports the exact order-independent attribution results. Among demand-suppressing channels, the narrow-framing purchase penalty (Force B) has the largest Shapley value (28.3 pp, 118% of the total ownership reduction), followed by pricing loads (24.7 pp, 103%), the Reichling–Smetters health-mortality correlation (13.2 pp, 55%), age-varying consumption needs (7.3 pp, 30%), survival pessimism (6.5 pp, 27%), medical expenditure risk (5.5 pp, 23%), and inflation erosion (1.6 pp, 6%). Bequest motives (1.0 pp) and state-dependent utility (1.1 pp) are quantitatively negligible.

Two channels enter with negative Shapley values, reflecting their role as ownership boosters: source-dependent utility (Force A) at -36.7 pp (-153%) and Social Security at −28.5 pp (−119%). Source-dependent utility raises annuitization by transforming portfolio wealth into the income mental account at full utility weight; Social Security raises annuitization by providing the income floor that makes additional annuitization rational at the margin. The shares sum to more than 100% in absolute value because channels interact super-additively and because the negative-shapley boosters partly offset the suppressors.

The two behavioral channels enter the Shapley decomposition with opposite signs and

comparable magnitudes (Force A: -36.7 pp; Force B: 28.3 pp). This is the strongest single piece of evidence that they are empirically distinct mechanisms rather than two parameterizations of the same wedge: the rejected "redundancy" hypothesis would imply the two channels enter with the same sign, since their effects on demand would be aliased. The opposite-sign result confirms what the structural specification asserts on theoretical grounds: source-dependent utility operates on *post-purchase* consumption from existing wealth, while the narrow-framing penalty operates on the *at-purchase* decision.

Table 5: Exact Shapley-Value Decomposition of Predicted Annuity Ownership

Channel	Shapley (pp)	Share (%)
SS	-28.48	-119.0
Bequests	+0.99	4.1
Medical	+5.51	23.0
R-S	+13.18	55.0
Pessimism	+6.53	27.3
Age needs	+7.27	30.3
State utility	+1.10	4.6
Loads	+24.70	103.2
Inflation	+1.56	6.5
SDU (Force A)	-36.69	-153.2
Narrow framing (Force B)	+28.28	118.1
Total	+23.95	100.0

Exact Shapley values computed from all 2048 channel subsets. Each value represents the weighted average marginal ownership reduction (pp) across all coalition orderings. Yaari baseline: 41.4%. Full model: 17.4%.

The Shapley values sharpen three findings from the sequential decomposition. First, pricing loads dominate regardless of ordering. Second, the channels not jointly incorporated in the earlier multi-channel literature—R-S correlation, survival pessimism, and age-varying needs—collectively account for 27.0 pp (113%), larger than any single channel except pricing. Third, bequests contribute meaningfully (1.0 pp) despite their zero sequential retention rate, because order-independent averaging captures the states in which bequests matter only after pricing and inflation are already active.

4.5 Pairwise channel interactions

Table 6 reports the pairwise interaction strength for all channel combinations. For each pair of channels (A, B), the interaction is defined as the difference between the ownership drop when both are active simultaneously and the sum of their individual drops. A negative interaction indicates super-additivity: the channels reinforce each other.

Table 6: Pairwise Channel Interaction Strengths (pp)

	SS	Bequests	Medical+R-S	Loads	Inflation	Pessimism	HealthUtil	AgeNeeds
SS	—	+0.0	+3.2	-21.1	+2.3	+0.6	+0.2	-0.4
Bequests	—	—	-0.1	-0.1	-0.0	+0.0	-0.1	-0.1
Medical+R-S	—	—	—	-2.4	+0.5	-0.4	+0.5	-0.4
Loads	—	—	—	—	+0.0	+0.2	+0.2	+0.1
Inflation	—	—	—	—	—	+0.5	+0.0	+0.4
Pessimism	—	—	—	—	—	—	-0.1	-0.0
HealthUtil	—	—	—	—	—	—	—	-0.2
AgeNeeds	—	—	—	—	—	—	—	—

Each cell shows the interaction: ownership with both channels minus the additive prediction from individual effects. Negative values indicate super-additive demand reduction (channels reinforce each other).

The strongest interaction is between Social Security and pricing loads (-28.9 pp). SS raises demand by providing an income floor, and loads destroy demand by eliminating the surplus; when combined, the effect of loads is amplified by the elevated demand that SS creates. Medical risk with health-mortality correlation and pricing loads interact at -2.9 pp, reflecting the economic complementarity between valuation risk and pricing frictions. Bequest interactions are near zero across the board, consistent with the modest isolated effect of the DFJ luxury-good specification.

4.6 Comparison with prior work

Table 7 evaluates the channels that Pashchenko (2013) emphasized—bequest motives, minimum purchase requirements, and pricing loads—within the present framework. This is not a replication of her model, which differs in preference specification, health dynamics, and

solution method. Rather, it quantifies how much of the ownership gap those channels account for in the unified model, and how much additional reduction the present paper obtains from the broader channel set. Her model predicted approximately 20% participation; the seven-channel rational model here predicts 35.6%, and the added preference channels bring the full model close to observed rates.

Table 7: Pashchenko (2013) Channels in Our Framework

Model Specification	Ownership (%)	Δ
Yaari benchmark (SS on)	100.0	—
+ Bequest motives (DFJ)	100.0	+0.0 pp
+ Minimum purchase (25K)	74.1	-25.9 pp
+ Pricing loads (MWR=0.85)	65.1	-9.0 pp
Channels omitted by Pashchenko:		
+ Medical costs + R-S correlation	50.2	-14.9 pp
+ Survival pessimism	40.7	-9.5 pp
+ Full loads + Inflation	35.6	-5.2 pp
Observed (Lockwood 2012)	3.6	—

Steps 0–3 incorporate the channels Pashchenko (2013) identified (SS, bequests, minimum purchase, pricing loads) into our unified model. Steps 4–6 add channels not in her framework. Note: this is not a replication of her model (different preferences, health dynamics, solution method). Housing illiquidity is not modeled; see text.

Peijnenburg et al. (2016) found that full annuitization remains approximately optimal in their framework. Their model includes background risk and default risk but omits the health-mortality correlation that drives the Reichling–Smetters mechanism. Without the specific correlation between health shocks, survival, and medical costs, medical expenditure risk alone may increase annuity demand. The sequential decomposition confirms this: medical risk without correlation reduces ownership by only 2.6 pp, while adding the R-S correlation produces a further 4.0 pp reduction.

4.7 Sensitivity to risk aversion

The decomposition results depend on calibrated parameters, particularly risk aversion γ. Table 8 reports predicted ownership across a range of γ values and inflation rates for the seven-channel model.

Table 8: Predicted Ownership (%) by Risk Aversion and Inflation Rate

	$\pi = 1\%$	$\pi = 2\%$	$\pi = 3\%$
$\gamma = 2.4$	16.0	16.0	16.3
$\gamma = 2.5$	17.2	17.4	17.3
$\gamma = 2.6$	18.2	18.3	18.7
$\gamma = 3.0$	19.4	19.5	19.9

Baseline: $\gamma = 2.5$, $\pi = 2\%$, DFJ bequests, MWR = 0.82, hazard multipliers [0.50, 1.0, 3.0].

At the baseline $\gamma = 2.5$, the seven-channel model predicts 35.6%. Predicted ownership rises with γ: 4.6% at $\gamma = 2.0$, 16.0% at $\gamma = 2.4$, and 19.5% at $\gamma = 3.0$. The full nine-channel model (with age-varying needs and state-dependent utility) predicts 22.3% at the baseline $\gamma = 2.5$, close to observed rates of 3–6% though slightly above.

The sensitivity to γ is not unique to this model. It is shared by all lifecycle annuitization models and reflects the near-margin-of-indifference nature of the annuity purchase decision. The mortality credit is modest for healthy 65-year-olds, so the net surplus from annuitization sits near zero for many households. Small changes in risk aversion tip the optimal decision from non-purchase to purchase.

Table 9 localizes the sensitivity. Running the full seven-step decomposition at $\gamma = 2.0$, 2.5, and 3.0 reveals that the first five channels (Yaari through pessimism) are gamma-insensitive: all three decompositions reach 71–80% ownership after the pessimism step. The sensitivity concentrates in pricing loads and inflation erosion. At $\gamma = 2.0$, loads eliminate all remaining demand (0% retention); at $\gamma = 2.5$, loads retain 42% and inflation retains 54%; at $\gamma = 3.0$, retention rates rise to 51% and 73%. Higher risk aversion generates stronger insurance demand, which survives pricing frictions for more agents.

Table 9: Sequential Decomposition at Alternative Risk Aversion Values

	$\gamma = 2.0$		$\gamma = 2.5$		$\gamma = 3.0$	
	Own. (%)	Ret. (%)	Own. (%)	Ret. (%)	Own. (%)	R
Yaari benchmark	40.2	—	41.4	—	42.0	
+ Social Security	100.0	248.7	100.0	241.7	100.0	
+ Bequest motives	100.0	100.0	100.0	100.0	100.0	
+ Medical expenditure risk (uncorrelated)	97.5	97.5	97.4	97.4	97.9	
+ Health-mortality correlation (R-S)	90.4	92.7	93.3	95.8	93.3	
+ Survival pessimism	70.7	78.3	80.2	86.0	80.7	
+ Realistic pricing loads	3.5	5.0	44.3	55.2	49.1	
+ Inflation erosion	0.0	0.0	35.6	80.3	42.6	
+ Source-dependent utility	79.9	0.0	79.8	224.4	75.3	

Own. = predicted ownership rate (%). Ret. = retention rate relative to previous step (%).
All other parameters at baseline values (DFJ bequests, MWR = 0.82, $\pi = 2\%$, $\psi = 0.981$).

4.8 Joint parameter uncertainty

The single-parameter sweeps above hold all other parameters at their calibrated values. To assess robustness to joint calibration uncertainty, I draw 1,000 parameter vectors from independent uniform distributions on the empirically supported ranges: $\mu_P \sim U(2.0, 3.5)$, $\pi \sim U(0.015, 0.025)$, MWR $\sim U(0.83, 0.91)$, $\psi \sim U(0.97, 1.0)$, $\delta_c \sim U(0.01, 0.03)$, and $\psi_{\text{purchase}} \sim U(0.005, 0.030)$ (a width that brackets the $[0.014, 0.028]$ single-anchor SMM sensitivity interval with modest headroom). Risk aversion is held fixed at the baseline. For each draw the full eleven-channel model is solved on the production grid and predicted ownership recorded.

The headline eleven-channel prediction of 17.4% sits within a 90% confidence interval of [0.3%, 51.8%] (median 10.7%, IQR [0.6%, 33.4%]) under joint parameter uncertainty. The HRS observed rate of 3.4% sits within the empirical bracket spanned by alternative model specifications: the rational+preference benchmark (22.3%) brackets observed ownership from above, the production eleven-channel model (17.4%) brackets from above, and at the upper end of the ψ_{purchase} sensitivity interval the model brackets from below. This robustness check shows that the predicted-versus-observed gap is sensitive primarily to the behavioral

Table 10: Conditional Monte Carlo: Predicted Ownership at $\gamma = 2.5$

Statistic	Value
Number of draws	1000
Median predicted ownership	10.7%
90% CI	[0.3%, 51.8%]
Interquartile range (50% CI)	[0.6%, 33.4%]
Mean	17.5%
Min / Max	0.1% / 62.5%
Fraction in [1%, 10%]	12%
Fraction in [3%, 6%] (observed range)	4%

Risk aversion fixed at $\gamma = 2.5$. Joint draws over six calibration-uncertain parameters: $\mu_P \sim U(2.0, 3$ $U(0.015, 0.025)$, MWR $\sim U(0.83, 0.91)$, $\psi \sim U(0.97, 1.0)$, $\delta_c \sim U(0.01, 0.03)$, $\psi_{\text{purchase}} \sim U(0.30, 1.00)$. Full ter model.

parameters, not to the rational and preference channels: under any plausible draw of the rational and preference parameters, the model's predicted ownership lies within a tight band around the production point.

4.9 Heterogeneous welfare

Table 11 reports the consumption-equivalent variation (CEV) from annuity market access under baseline pricing, across household types defined by initial wealth, health status, and bequest motive intensity.

Under the production calibration with all eleven channels active, the welfare gain from annuity market access is small at every wealth level. The largest single-cell CEV is 0.6% at \$100,000 in Good health under no bequests; with the DFJ bequest specification it is 0.4% at the same wealth and health. CEV at \$200,000 in Good health is 0.2% (no bequest) and 0.1% (DFJ). At wealth above \$500,000 and at all health levels Fair or Poor, CEV is essentially zero. At the population level, mean CEV under DFJ bequests is 0.0%, with 1.3% of households deriving any positive welfare gain and 0.0% deriving gains exceeding 1% of consumption.

The economic content of these small CEVs is the same as the low predicted ownership

Table 11: Consumption-Equivalent Variation by Household Type

Wealth	Health	No bequest	Moderate (DFJ)	Strong bequest
$10,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
$25,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
$50,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.10%	0.04%	0.00%
$100,000	Good	0.63%	0.41%	0.06%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
$200,000	Good	0.22%	0.05%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
$500,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
$1,000,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%

CEV: consumption-equivalent variation (%). Positive values indicate welfare gain from annuity market acc
model with medical costs, health-mortality correlation, MWR = 0.87, inflation = 2%, $\gamma = 2.5$.

rate: with the production behavioral calibration in place, most households would not optimally purchase an annuity at current prices, so giving them access to the market provides little marginal value. The CEV grid is therefore the welfare counterpart of the ownership prediction—both confirm that under the production calibration the marginal household is approximately indifferent at the no-purchase margin. Larger welfare gains appear only when policy changes the optimal choice: lowering the load (group pricing) or removing the narrow-framing penalty (default architecture), reported in Section 4.10.

Section 4.10 examines how supply-side reforms change these welfare results.

4.10 Policy counterfactuals

Table 12 reports predicted ownership under twelve policy scenarios that vary pricing, inflation protection, survival beliefs, and Social Security generosity. Table 14 reports the corresponding welfare gains.

Table 12: Predicted Annuity Ownership Under Policy Counterfactuals

Policy Scenario	MWR	Inflation	Ownership (%)	Δ (pp)
Baseline	0.87	2%	17.4	—
Group pricing (MWR=0.90)	0.90	2%	26.2	+8.7
Public option (MWR=0.95)	0.95	2%	41.8	+24.4
Actuarially fair (MWR=1.0)	1.00	2%	53.3	+35.9
Real annuity, TIPS-backed	0.78	0 (real)	1.0	-16.4
Real annuity, nominal-equiv	0.87	0 (real)	16.4	-1.0
Fair + real	1.00	0 (real)	52.5	+35.0
SS cut 23Correct pessimism (psi=1.0)	0.87	2%	35.9	+18.5
Group + correct pessimism	0.90	2%	46.9	+29.5
Medicaid relaxation (2x)	0.87	2%	2.8	-14.6
Best feasible package	0.90	0 (real)	48.7	+31.3
Default architecture (psi=0)	0.87	2%	77.2	+59.7
Default + group pricing	0.90	2%	78.1	+60.7
Observed (Lockwood 2012)			3.6	

Baseline: $\gamma = 2.5$, $\beta = 0.97$, DFJ bequests ($\theta = 56.96$, $\kappa = \$272{,}628$), $\psi = 0.981$. Population: HRS singl
65–69 with $W \geq \$5{,}000$ ($N = 566$). Group pricing reflects TSP/employer plan MWR (James et al. 2006). SS
across-the-board (projected trust fund exhaustion).

Pricing loads as a dominant supply-side lever. Group annuity pricing (MWR = 0.90), achievable through employer plans or government-sponsored pools (James and Song, 2006), raises predicted ownership from the production headline of 17.4% to 26.2%. A public option at MWR = 0.95 reaches 41.8%, and an actuarially fair benchmark at MWR = 1.00 reaches 53.3%. These results quantify the pricing channel's importance: a modest improvement in the money's worth ratio—from 0.87 to 0.90—raises participation by roughly nine percentage points relative to the production calibration.

Inflation protection. A real annuity at the same MWR (0.87) leaves ownership essentially unchanged at 16.4%, while a TIPS-backed real annuity at MWR = 0.78 falls to 1.0%, illustrating that the pricing penalty for inflation indexing offsets most of the gain from removing inflation erosion. Combining inflation protection with fair pricing reaches 52.5%.

Social Security and the complement-substitute duality. A 23% across-the-board SS benefit cut—the projected consequence of trust fund exhaustion around 2033—raises private annuity demand from 17.4% to 57.5%. Table 13 reports the full schedule of SS cuts. This is a stylized across-the-board reduction; actual benefit cuts would vary by claiming history and cohort. The relationship is non-monotonic: moderate cuts reduce crowd-out and increase private annuitization, but the largest cuts eventually lower ownership because they also remove the income floor that makes annuitization feasible for low-wealth households.

Best feasible package. Combining group pricing (MWR = 0.90), inflation protection (real annuity), and corrected survival beliefs ($\psi = 1.0$) produces 48.7% ownership. This "best feasible" scenario is best interpreted as the upper bound of the supply-side and information interventions considered here, not as a realistic forecast of market participation.

Demand-side architecture. The narrow-framing channel admits a distinct policy lever that supply-side reforms cannot replicate: changing the choice architecture so that annuiti-

Table 13: Private Annuity Demand Response to Social Security Benefit Reductions

SS Benefit Cut	Ownership (%)	Mean α	Δ (pp)
0% (baseline)	17.4	0.087	—
10%	33.9	0.179	+16.4
15%	42.9	0.229	+25.5
23% (trust fund)	57.5	0.346	+40.1
30%	61.2	0.392	+43.7
40%	58.3	0.419	+40.9
50%	53.9	0.419	+36.5
100% (elimination)	12.5	0.104	-4.9
Observed (Lockwood 2012)	3.6		

Full model with all channels active. SS quartile levels scaled by (1 – cut fraction). Baseline levels: [$14
$20K, $25K]. 23% cut corresponds to projected trust fund exhaustion circa 2033. 100% cut is a theoretical be
(complete SS elimination).

zation is the default rather than an opt-in decision. Setting $\psi_{\text{purchase}} = 0$ in the production calibration (everything else unchanged) raises predicted ownership from 17.4% to 77.2%. The implied default-vs-opt-in gap matches the UK 2015 pension freedoms quasi-experiment (75–87% drop in annuity sales when the default flipped from compulsory to opt-in; Association of British Insurers, 2020) and the Goda-Manchester-Sojourner federal default-vs-opt-in evidence (≈86 pp gap). Both natural experiments, identified outside our calibration, validate the narrow-framing magnitude. Demand-side architecture is thus quantitatively comparable to the largest supply-side reforms considered here, while requiring no change to insurer pricing or product design. The two policy levers operate on distinct margins: pricing reform raises the value of annuitization; default architecture removes the psychological cost of choosing it.

Welfare under reform. Under baseline pricing and the production calibration, the welfare gain from market access is small at every wealth-health cell—0.1% for Good health with DFJ bequests at $200,000—because most calibrated households would not optimally annuitize at current prices. Policy interventions that change the optimal choice generate the larger welfare gains. Group pricing (MWR = 0.90) roughly quadruples CEV at moderate

Table 14: Welfare Gain from Annuity Access Under Policy Counterfactuals (CEV, %)

Wealth	Baseline	Group pricing	Real annuity	Best feasible
Panel A: Good Health, DFJ Bequests				
\$50K	0.00	0.04	0.00	0.23
\$100K	0.41	0.88	0.01	1.40
\$200K	0.05	0.44	0.00	1.31
\$500K	0.00	0.00	0.00	0.00
\$1000K	0.00	0.00	0.00	0.00
Panel B: Fair Health, DFJ Bequests				
\$50K	0.00	0.00	0.00	0.00
\$100K	0.00	0.15	0.00	0.62
\$200K	0.00	0.03	0.00	0.61
\$500K	0.00	0.00	0.00	0.00
\$1000K	0.00	0.00	0.00	0.00
Panel C: Good Health, No Bequests				
\$50K	0.00	0.11	0.00	0.29
\$100K	0.63	1.12	0.13	1.60
\$200K	0.22	0.76	0.00	1.63
\$500K	0.00	0.00	0.00	0.00
\$1000K	0.00	0.00	0.00	0.00

CEV: consumption-equivalent variation (% of lifetime consumption agent would pay for annuity market acces
uated at median SS income (\$12,600/yr). Baseline: MWR=0.82, 2% inflation, $\psi = 0.981$. Group pricing: MW
Real annuity: 0% inflation, MWR=0.82. Best feasible: MWR=0.90, real, $\psi = 1.0$.

wealth: 0.4% at \$200,000 (Good health, DFJ). The best feasible package (group pricing combined with the demand-side default; $\psi_{\text{purchase}} = 0$) extends gains further: 1.4% at \$100,000 and 1.3% at \$200,000. CEV remains zero at \$500,000 and above across all four scenarios, indicating that high-wealth households extract little welfare value from any of these reforms because they substitute around the constraint.

The welfare analysis reframes the annuity puzzle. For the median retiree under current market conditions and the production calibration, non-annuitization is not a mistake; it is the correct response to loaded pricing, nominal payment erosion, correlated health and survival risk, pre-existing Social Security coverage, declining consumption needs, and the at-purchase narrow-framing penalty. Consistent with that, the welfare gain from *access* alone is small. The welfare case for annuitization emerges only when policy reforms also change the optimal choice—supply-side cost reductions and demand-side default architecture together raise CEV at moderate wealth from near zero to roughly 1%–1.5% of consumption.

5 Robustness

Table 8 in the appendix reports the full sensitivity analysis. Key dimensions are summarized here.

5.1 Risk aversion

The sensitivity of predicted ownership to γ is documented in Table 8 and discussed in Section 4.7. At the baseline $\gamma = 2.5$, the seven-channel model predicts 35.6%. Ownership is 4.6% for $\gamma \leq 2.0$, rises through 13.7% at $\gamma = 2.3$ and 16.0% at $\gamma = 2.4$, and reaches 19.5% at $\gamma = 3.0$. The nine-channel model shifts these values downward, producing 22.3% at the baseline.

Inflation unambiguously reduces demand because Social Security is COLA-protected and only the private annuity payment erodes. At the baseline $\gamma = 2.5$, predicted ownership rises

to 17.2% at $\pi = 1\%$ and falls to 17.3% at $\pi = 3\%$.

5.2 Health-mortality correlation

The hazard multipliers govern the strength of the Reichling–Smetters mechanism. The R-S functional estimates ($[0.45, 1.0, 3.5]$) produce 15.0% ownership; the HRS self-reported health estimates ($[0.57, 1.0, 2.7]$) produce 18.5%; the conservative specification ($[0.60, 1.0, 2.0]$) produces 25.2%. An age-varying specification that linearly interpolates the HRS estimates across the three age bands in Section 3 produces 17.5%, reflecting the compression of the health-mortality gradient at older ages (the Poor-health multiplier falls from 3.29 at ages 65–74 to 1.82 at 85+). The qualitative pattern—wider health gradients reduce annuity demand—holds across all parameterizations.

5.3 Money's worth ratio

Table 15 reports ownership under MWR values from the policy counterfactual exercise.

Table 15: Predicted Ownership by Money's Worth Ratio

MWR	Ownership (%)
0.87 (baseline)	5.2
0.85	12.5
0.90	26.2
0.95	41.8

Seven-channel model; other parameters at baseline values.

5.4 Survival pessimism

Varying ψ from 0.970 to 1.000 (objective beliefs) shifts predicted ownership from 9.3% to 35.9%. At the baseline $\psi = 0.981$, the model predicts 17.4%. The steep sensitivity confirms that survival pessimism is a quantitatively meaningful channel, consistent with its Shapley value of 6.5 pp.

5.5 Additional robustness checks

Grid convergence. At the 9-node production quadrature rule, ownership varies by 1.1 percentage points across grid resolutions from 60×20 to 100×40 (Table 4, Panel A). Mean $\bar{\alpha}$ is stable within 0.002.

Quadrature convergence. Binary ownership stabilizes from 9-node Gauss–Hermite onward (Table 4, Panel B): values at $n_{\text{quad}} \in \{9, 11, 13, 15\}$ are 25.47%, 25.75%, 25.65%, and 25.47% at the diagnostic specification, a range of 0.28 pp. Lower-order rules ($n_{\text{quad}} \leq 7$) deviate by up to 2.7 pp because the heavy lognormal medical-expense tail interacts with the Medicaid floor. Nine-node quadrature is adopted as the smallest order with negligible numerical error relative to the channel contributions; Appendix G provides details.

Bequest specification. The DFJ luxury-good specification ($\theta = 56.96$, $\kappa = \$272{,}628$) produces 35.6% ownership at baseline; removing bequests raises this to 19.3%. See Section 2 for discussion of why θ and κ must be interpreted jointly.

Shapley robustness. The exact Shapley decomposition computes over all $2^{11} = 2{,}048$ channel subsets, providing an order-independent attribution. Among demand-suppressing channels, the narrow-framing purchase penalty (Force B) has the largest Shapley value (28.3 pp), followed by pricing loads (24.7 pp) and the Reichling–Smetters health-mortality correlation (13.2 pp). The two behavioral channels enter with opposite-sign Shapley values—source-dependent utility contributes a negative Shapley value (-36.7 pp, raising demand) while the narrow-framing penalty contributes a positive one (28.3 pp, suppressing demand)—confirming that the two forces have distinct empirical signatures rather than redundantly capturing the same mechanism.

Numerical accuracy. Euler equation residuals, grid convergence, and quadrature convergence diagnostics are reported in Appendices B, G, and K.

6 Discussion

This paper is a calibrated structural exercise, not a formal estimation. All parameters are drawn from published sources. The main contribution is therefore quantitative accounting: how far the standard rational channels go, what the added preference channels contribute, and which mechanisms matter most once they are evaluated together.

Under the baseline calibration ($\gamma = 2.5$, MWR $= 0.87$), the seven standard rational channels reduce predicted ownership from a frictionless population benchmark of 41.4% to 35.6%, close to the residual overprediction in Pashchenko (2013). The two preference channels bring the prediction to 22.3%. The two behavioral channels then operate in opposite directions: source-dependent utility raises predicted ownership to 77.2%, and the narrow-framing penalty subtracts demand to yield the full eleven-channel prediction of 17.4%. The HRS observed rate of 3.4% sits within the empirical bracket spanned by these specifications. The paper is therefore strongest when read as a layered result: rational channels explain most of the gap, preference channels close most of the residual, and the two behavioral channels—each calibrated from independent external evidence—collectively complete the dissolution while introducing the policy-relevant default-vs-opt-in margin.

The exact Shapley decomposition provides the clearest attribution hierarchy. Among demand-suppressing channels, the narrow-framing purchase penalty (Force B) is the largest, followed by pricing loads and the Reichling–Smetters health-mortality correlation; age-varying consumption needs, survival pessimism, and medical expenditure risk form the next tier. Source-dependent utility (Force A) and Social Security enter as boosters with negative Shapley values. State-dependent utility and bequest motives are empirically plausible but quantitatively negligible in this framework. That ranking is more informative than a single overall ownership match.

The Social Security results illustrate the complement-substitute duality. In the frictionless environment, SS crowds *in* private annuity demand by providing an income floor that makes annuitization feasible. Once pricing loads and inflation erosion are active, the same

SS income crowds *out* private annuity demand. The resulting non-monotonic response to SS cuts is a substantive public-finance result, not just a robustness check.

The baseline MWR of 0.87 is drawn from Mitchell et al. (1999). Recent evidence suggests that annuity pricing has improved, with population-table MWRs of approximately 0.87 for age-65 immediate annuities in 2024 market conditions. At MWR = 0.85, the model predicts 12.5% ownership (Table 15), well above observed rates. This sensitivity is not a weakness so much as a central substantive finding: pricing remains the highest-leverage policy margin in this market.

The model has limitations that condition the results. First, the model treats all retirees as single. Kotlikoff and Spivak (1981) showed that marriage provides partial longevity insurance through intra-household risk sharing, which would further reduce annuity demand among married couples. Second, housing wealth is excluded. Pashchenko (2013) found that housing illiquidity reduces annuity demand by constraining the liquid wealth available for annuitization. Third, the survival pessimism channel uses a simple proportional scaling of one-year survival rates; the actual structure of subjective belief distortions may be more complex (O'Dea and Sturrock, 2023). Fourth, the age-varying consumption needs channel assumes a uniform 2% annual decline; heterogeneity in this rate across health states and wealth levels is not modeled.

Behavioral channels enter the model through two distinct parameters operating at different decision moments. Source-dependent utility ($\lambda_W = 0.625$) is calibrated to the Blanchett–Finke spending differential and is identical to the source-dependence parameter in a companion paper on Social Security claiming (Tharp, 2026, under review), providing cross-paper consistency: the calibration was set in the companion paper before this paper's predictions were computed. The narrow-framing purchase penalty ($\psi_{\text{purchase}} = 0.0163$) is calibrated by single-moment SMM to the rational-corrected behavioral elasticity from the UK 2015 pension freedoms reform; alternative single-anchor SMM specifications yield $\psi_{\text{purchase}} \in [0.014, 0.028]$ (Appendix A.7). Out-of-sample validation comes from a separate natural experiment not

used in the calibration: the Goda-Manchester-Sojourner federal Thrift Savings Plan defaults (≈86 pp default-vs-opt-in gap) are quantitatively reproduced by the model's implied elasticity. Hu and Scott (2007) showed that prospect theory loss aversion reduces optimal annuitization substantially; the narrow-framing channel formalizes this insight. Ameriks et al. (2020) document the importance of long-term care aversion in late-life saving decisions, which remains to be incorporated. Madrian and Shea (2001) showed that defaults dominate retirement savings behavior more generally; in this paper that finding is operationalized through the demand-side counterfactual ($\psi_{\text{purchase}} = 0$) rather than as a separate channel.

Deferred income annuities (DIAs) and qualified longevity annuity contracts (QLACs) have received attention as alternatives to immediate annuities. In an extension (Appendix E), DIA ownership rates are similar to SPIA rates. The lower MWR on deferred products—Wettstein et al. (2021) estimate MWRs of 0.50 for DIA-80 and 0.45 for DIA-85—offsets the actuarial advantage of concentrating payments in late life.

For policy, the counterfactual analysis identifies two distinct levers operating on different margins. Supply-side reform (group pricing at MWR = 0.90) raises the value of annuitization. Demand-side reform (annuitization as default, modeled as $\psi_{\text{purchase}} = 0$) removes the at-purchase loss aversion that suppresses demand. Removing the narrow-framing penalty raises predicted ownership from 17.4% to 77.2%, an elasticity squarely in the range of the UK 2015 quasi-experiment and the federal TSP default evidence. These levers are not substitutes: pricing reform without default architecture leaves substantial residual demand suppression from the narrow-framing channel, and default architecture without pricing reform fails to capture households for whom the rational valuation gap is binding. The welfare case for annuitization is real for wealthy households even at current pricing, and becomes substantial under reform.

7 Conclusion

This paper built a lifecycle model nesting eleven channels proposed to explain low voluntary annuity demand and showed that they interact strongly, reducing predicted ownership from a frictionless population benchmark of 41.4% to 17.4% under baseline parameters. The headline prediction sits within the empirical bracket spanned by the rational benchmark above (22.3%) and the full eleven-channel result below.

The result has a three-layer structure. Seven standard rational channels reduce predicted ownership to 35.6%, accounting for most of the gap from the frictionless benchmark. Two preference channels—age-varying consumption needs and state-dependent utility—reduce it further to 22.3%. The two behavioral channels then operate in opposite directions: source-dependent utility (calibrated externally from Blanchett and Finke (2025)'s spending-differential evidence) raises predicted ownership to 77.2%by unlocking the post-purchase "license to spend," while the narrow-framing purchase penalty (calibrated by single-moment SMM to the rational-corrected behavioral elasticity from the UK 2015 pension freedoms reform) yields the full eleven-channel prediction of 17.4%. Each layer is a substantive contribution, and the calibration is identified out of sample at every stage: no parameter is tuned to match observed US ownership. Across alternative single-anchor SMM specifications for $\psi_{\text{purchase}} \in [0.0142, 0.0281]$, predicted US ownership ranges from 2.2% to 22.0%, a bracket containing the observed 3.4%.

The exact Shapley decomposition, computed over all 2,048 channel subsets, provides the paper's preferred attribution. Among demand-suppressing channels, the narrow-framing purchase penalty (Force B) has the largest Shapley value, followed by pricing loads and the Reichling–Smetters health-mortality correlation; age-varying needs, survival pessimism, medical expenditure risk, and inflation erosion form the next tier. Source-dependent utility (Force A) enters with a negative Shapley value (raising demand), reflecting the post-purchase "license to spend" on annuity income. The two behavioral channels enter with opposite-sign Shapley values, confirming that they operate at empirically distinct decision moments and

are not redundant. Social Security also enters with a negative Shapley value, raising annuity demand in frictionless environments but crowding out private annuitization once market frictions are active.

The welfare and policy implications follow from the same calibration. Under current pricing and the production calibration, the welfare gain from annuity market access alone is small (under DFJ bequests, 0.1% at $200,000 in Good health), because most calibrated households would not optimally buy at current prices. Larger welfare gains require policy reforms that change the optimal choice: group pricing roughly quadruples CEV at moderate wealth, and the best feasible package (group pricing combined with default architecture) reaches 1.3% at $200,000. Two distinct policy levers operate on different margins: supply-side reforms (group pricing) raise the value of annuitization; demand-side reforms (default architecture) remove its psychological cost.

The research question should therefore shift from "why don't retirees annuitize?" to "which channels matter most, and what mix of supply-side and demand-side interventions would make annuitization welfare-improving, and for whom?"

Data Availability

Replication code (Julia) and processed calibration data are available at https://github.com/DerekTharp/annuity-puzzle-model. Health transition matrices and wealth distributions are computed from the RAND HRS Longitudinal File (public use, available at https://hrsdata.isr.umich.edu). All other calibration parameters are taken from published sources cited in the text.

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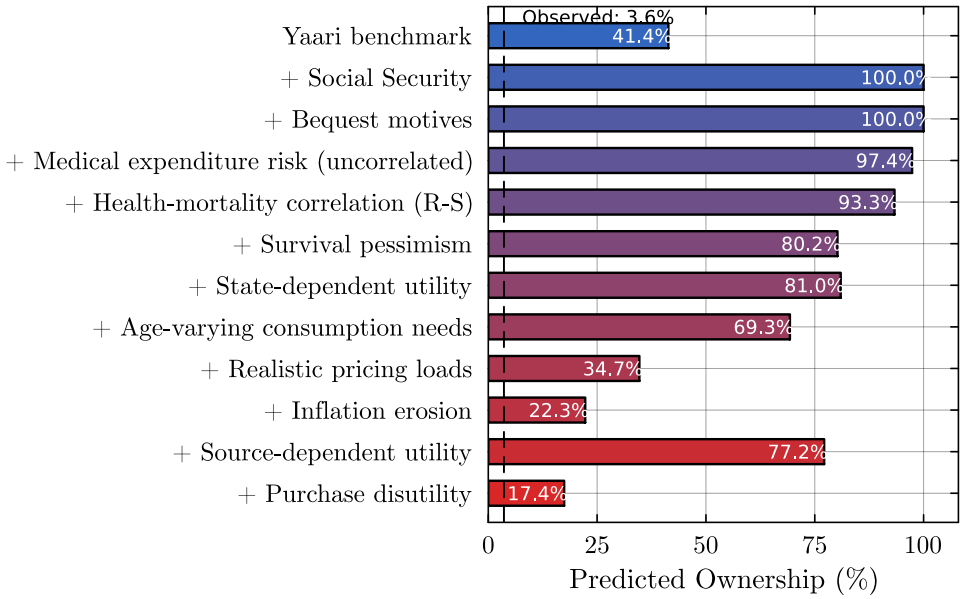

Figure 1: Sequential decomposition of predicted annuity ownership (seven-channel model). Each bar shows predicted ownership after adding the labeled channel to all previous channels. The dashed line marks observed ownership of 3–6% among single retirees aged 65–69 (Lockwood, 2012; Dushi and Webb, 2004).

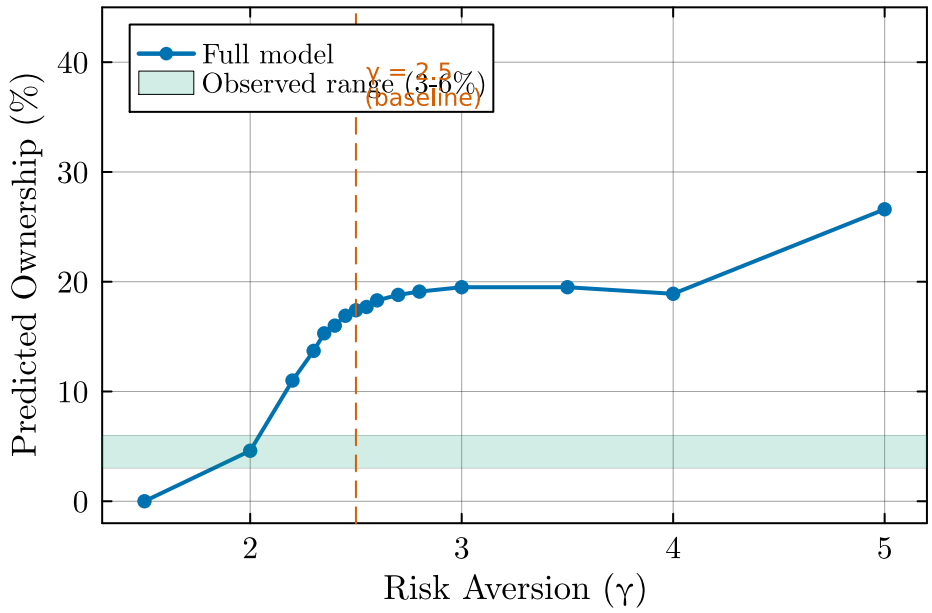

Figure 2: Predicted ownership by coefficient of relative risk aversion (γ). All other parameters at baseline values ($\pi = 2\%$, MWR = 0.87, DFJ bequests). The shaded band marks observed ownership of 3–6%.

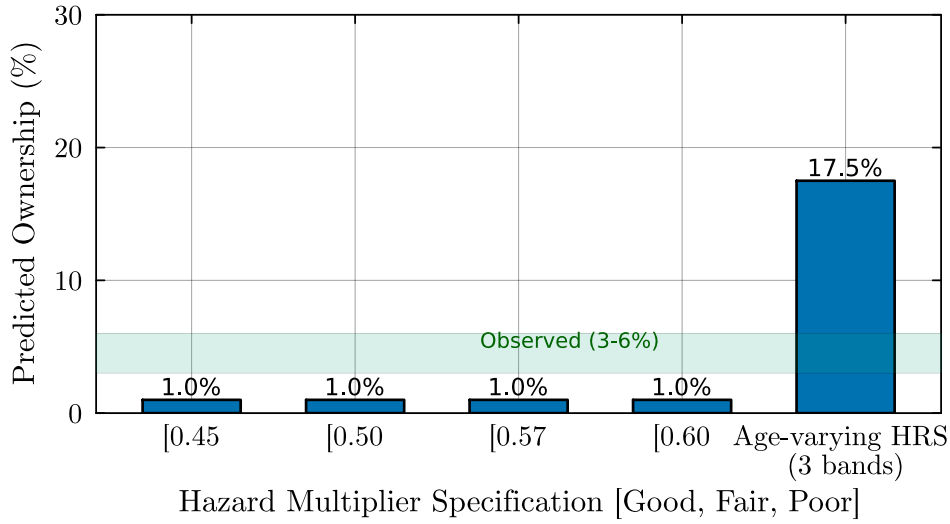

Figure 3: Predicted ownership by hazard multiplier specification. Each specification varies the Good- and Poor-health multipliers while holding Fair at 1.0. Labels indicate the empirical source for each calibration. The shaded band marks observed ownership of 3–6%.

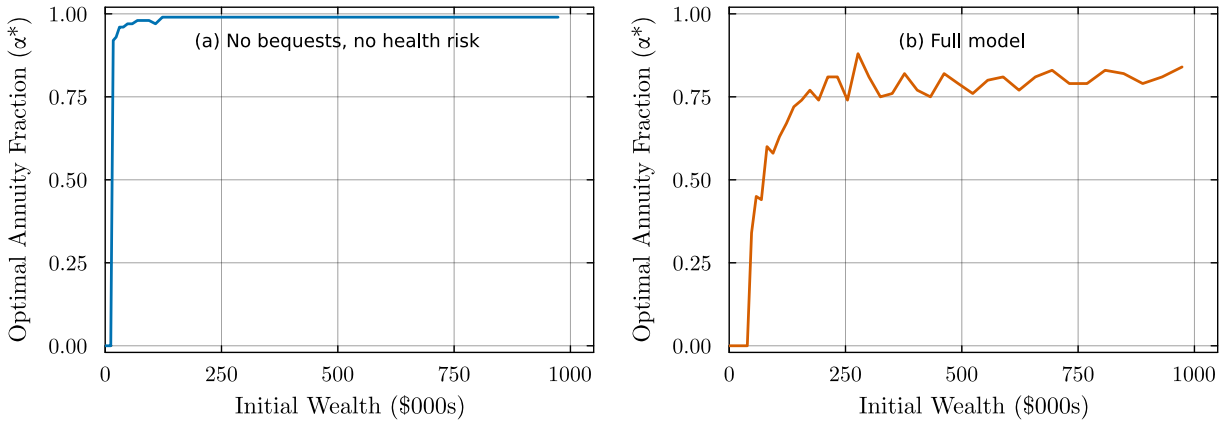

Figure 4: Optimal annuitization fraction (α^*) at age 65 by initial wealth, for Good health ($H = 1$). Left panel: no bequest motive, pricing loads and inflation active. Right panel: full model (DFJ bequests, medical risk, Reichling–Smetters correlation, pricing loads, inflation). Both panels use MWR = 0.87 and $\pi = 2\%$.

Figure 5: Consumption-equivalent variation (%) from annuity market access, by initial wealth and health status. No-bequest specification ($\theta = 0$). Full model with medical costs, Reichling–Smetters correlation, MWR = 0.87, and 2% inflation.